Raj Gandhi

Curriculum vitae

Malott Hall, Cornell University
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Research interests

I am interested in applying formal group laws and tools from generalized cohomology theories (e.g. algebraic cobordism, connective K-theory, and their equivariant counterparts) to study problems in Schubert calculus, the geometry of matroids, and torus-equivariant degenerations of varieties.

Education

Sep. 2021 - Aug. 2027 **Ph.D. in Mathematics**, *Cornell University*
Thesis title: *TBA.*
Advisor: Allen Knutson.

Sep. 2019 - Aug. 2021 **M.Sc. in Mathematics**, *University of Ottawa*
Thesis title: *Oriented cohomology rings of the semisimple linear algebraic groups of ranks* 1 *and* 2. *ruor.uottawa.ca/handle/10393/42566*.
Advisors: Alistair Savage and Kirill Zainoulline.

2015–2019 **B.Sc. in Physics-Mathematics**, *University of Ottawa*

Research

Preprints/Publications

1. **Gandhi, R.** A technique for computing oriented cohomology rings of semisimple algebraic groups. Preprint (2026). `https://arxiv.org/abs/2607.07900`
2. **Gandhi, R.**, Partida, E. Algebraic cobordism rings of wonderful varieties and matroids. Preprint (2026). `https://arxiv.org/abs/2606.12645v1`
3. **Gandhi, R.** Motivic Segre classes of Schubert cells and the connective formal group law. Preprint (2026). `https://arxiv.org/abs/2605.26556`
4. **Gandhi, R.** The formal affine Demazure algebra and real finite reflection groups. *Algebras and Representation Theory*. (2023). Vol. 26. No. 5. `https://link.springer.com/article/10.1007/s10468-022-10158-x`.
5. **Gandhi, R.**, Savage, A., Zainoulline, K. Diagrammatics for F_4. *Bulletin of the London Mathematical Society.* (2023). Vol. 55. No. 1. `https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/blms.12733`.
6. **Gandhi, R.** Decomposing Frobenius Heisenberg categories, *Journal of Algebra and its Applications*. (2020). Vol. 19. No. 5. 31 pages. `doi.org/10.1142/S0219498820500942`.

Refereed Conference Proceedings

1. **Gandhi, R.**, Partida, E. Algebraic cobordism rings of wonderful varieties and matroids. *Séminaire Lotharingien de Combinatoire.,* To appear. https://sites.math.washington.edu/fpsac2026/public/abstracts/gandhi.pdf.
2. **Gandhi, R.** Deforming the motivic Segre classes of Schubert cells in the Grassmannian. *Séminaire Lotharingien de Combinatoire.* (2025). Issue 93B. https://www.mat.univie.ac.at/~slc/wpapers/FPSAC2025/12.pdf.
3. **Gandhi, R.**, Udell, G. A degeneration of the brick variety and a mixed subdivision of the associahedron into cubes. *Séminaire Lotharingien de Combinatoire.* (2025). Issue 93B. https://www.mat.univie.ac.at/~slc/wpapers/FPSAC2025/91.pdf.

Teaching Experience

Cornell University

- MATH 1910: Calculus for Engineers. Grader and recitation instructor. (Fall 2026).
- MATH 1110: Calculus I. Instructor. (Spring 2026).
- MATH 1920: Multivariable Calculus for Engineers. Grader and recitation instructor. (Fall 2025).

University of Ottawa

- MAT 1320X: Calculus I. Recitation instructor. (Summer 2021).
- MAT 2355: Introduction to Geometry. Grader. (Winter 2021).
- MAT 1348: Discrete Mathematics for Computing. Grader and recitation instructor.
  (Winter 2021).
- MAT 2141: Honours Linear Algebra. Grader. (Fall 2020).
- MAT 1341: Introduction to Linear Algebra. Grader and recitation instructor. (Fall 2020).
- MAT 1332: Calculus II for the Life Sciences. Grader and recitation instructor. (Fall 2020).
- MAT 1300X: Mathematical Methods I. Recitation instructor. (Summer 2020)
- MAT 1320: Calculus I. Grader and recitation instructor. (Winter 2020).
- MAT 1348: Discrete Mathematics for Computing. Grader and recitation instructor.
  (Winter 2020).
- MAT 1362: Mathematical Reasoning and Proofs. Grader. (Winter 2020).
- MAT 1362: Mathematical Reasoning and Proofs. Grader and recitation instructor. (Fall 2019).

Outreach and service

Canada/USA Mathcamp

June-August 2023 I was a **mentor** at Canada/USA Mathcamp. I taught five one-week-long courses: Knot invariants; Representation theory of the symmetric groups; Linear algebra through knots; Quiver representations II; Introduction to Schubert calculus.

Cornell Prison Education Program

Jan.-Feb. 2025 I was an **instructor** for MATH 112, a course offered at Auburn Correctional Facility through Cayuga Community College. (This course was canceled partway through due to a New York State prison guard strike.)

Sep.-Dec. 2024 I was a **tutor** for MATH 112, a course offered at Auburn Correctional Facility through Cayuga Community College.

Ithaca high school math seminar

Nov.-Dec. 2024 I taught a 16-lecture course at Ithaca high school. The topic was Tropical Geometry, and the problem sets for the course are here: `https://drive.google.com/drive/folders/1cqaSJSWWGK_gcvJ8pGHMyONpy520fwua?usp=sharing`

Jan.-March 2024 I taught a 16-lecture course at Ithaca high school. The topic was Partially Ordered Sets, and the notes for the course are here: `https://drive.google.com/drive/folders/1Do59gETOj4nmm8ylIAGWmUz480FXHOHH?usp=sharing`

Jan.-March 2023 I taught a 16-lecture course at Ithaca high school. The topic was Young Tableaux, and the notes for the course are here: `https://drive.google.com/file/d/1UdUDlT_izXgFtyYTwd70g6Ff7zScqt08/view?usp=sharing`

Cornell GRASSHOPR Program

March 2025 Anna Barth and I taught a three-lesson-long mini-course on the topic of cryptography to middle school students at Lehman Alternative Community School in Ithaca, NY.

March 2023 I taught a five-lesson-long mini-course on graph theory to middle school students at Lehman Alternative Community School in Ithaca, NY.

March 2022 Aria Beaupré and I taught a four-lesson-long mini-course on the topic of voting theory to middle school students at Namasté Montessori school in Ithaca, NY.

Cornell directed reading program

Fall 2025 My undergraduate mentees and I read “Matroids: A Geometric Introduction" by Gary Gordan and Jennifer McNulty.

Spring 2023 My undergraduate mentee and I read the first six chapters of “The Knot Book" by Colin C. Adams.

Fall 2022 My undergraduate mentee and I read the first three chapters of “Arithmetic of Elliptic Curves" by Joseph H. Silverman.

Presentations

Invited/Contributed Talks

May 2, 2026 **Algebraic Geometry Northeastern Series (AGNES)**, *Algebraic cobordism rings of wonderful varieties and matroids*, Rutgers University, (5 minute talk)

Feb. 16, 2026 **Informal mathematical physics seminar**, *Motivic Segre classes of Schubert cells and the connective formal group law*, Columbia University.

Feb. 12, 2026 **Geometry, Physics, and Representation Theory seminar**, *Motivic Segre classes of Schubert cells and the connective formal group law*, Northeastern University.

Jan. 29, 2026 **Algebra/Topology seminar**, *Motivic Segre classes of Schubert cells and the connective formal group law*, University of Albany.

June 26, 2025 **Algebra seminar**, *Deforming the motivic Segre classes of Schubert cells in the Grassmannian*, University of Ottawa.

May 1, 2025 **Combinatorics seminar**, *Deforming the motivic Segre classes of Schubert cells in the Grassmannian*, University of Toronto.

March 17, 2024 **Graduate Student Combinatorics Conference (GSCC)**, *Degenerating brick manifolds and subdividing the associahedron*, Carnegie Mellon University.

June 3, 2021 **Algebra and Geometry of Homogeneous Spaces**, *The formal group ring and real finite reflection groups*, University of Ottawa.

Posters

July 14, 2026 **Formal power series and algebraic combinatorics (FPSAC)**, *Algebraic cobordism rings of wonderful varieties and matroids*, University of Washington, Poster.

May 27, '26 **Affine combinatorics and quantum K-theory**, *Motivic Segre classes of Schubert cells and the connective formal group law*, Ohio State University, Poster.

Jan. 24, '26 **Combinatorial algebra meets algebraic combinatorics (CAAC)**, *Deforming the motivic Segre classes of Schubert cells in the Grassmannian*, Dalhousie University, Poster.

Nov. 22, '25 **Algebraic geometry northeastern series (AGNES)**, *Deforming the motivic Segre classes of Schubert cells in the Grassmannian*, University of Massachusetts Amherst, Poster.

July 21, '25 **Formal power series and algebraic combinatorics (FPSAC)**, *Deforming the motivic Segre classes of Schubert cells in the Grassmannian*, Hokkaido University, Poster.

March 22, 2025 **Mid-Atlantic Algebra, Geometry, and Combinatorics (MAAGC) Workshop**, *Deforming the motivic Segre classes of Schubert cells in the Grassmannian*, CUNY Graduate Center in New York, Poster.

June 26, 2024 **Encuentro Colombiano de Combinatoria (ECCO)**, *Generalized cohomology rings of rank 2 root systems*, Universidad del Cauca, Poster.

Undergraduate Posters

2016 The ring of twisted differential operators of reflection group $I_2(5)$. University of Ottawa library. `ruor.uottawa.ca/handle/10393/36283`.

Informal talks

Feb. 20, '24 **Olivetti club**, *Linear algebra through quiver representations*, Cornell University

May 2, '23 **Olivetti club**, *String calculus: a calculus with a twist*, Cornell University

Conferences and workshops

Attended

July 13-17, '26 **Formal power series and algebraic combinatorics (FPSAC)**, *University of Washington*, Seattle, Washington

May 26-29, '26 **Affine combinatorics and quantum K-theory**, *Ohio State University*, Columbus, OH

May 1-3, '26 **Algebraic Geometry Northeastern Series (AGNES)**, *Rutgers University*, New Brunswick, NJ

Jan. 23-25, '26 **Combinatorial algebra meets algebraic combinatorics (CAAC)**, *Dalhousie University*, Halifax, Nova Scotia

Nov. 21-23, '25 **Algebraic geometry northeastern series (AGNES)**, *University of Massachusetts Amherst*, Amherst, Massachusetts

Aug. 18-22 '25 **Workshop on algebra and representation theory**, *University of Oregon*, Eugene, Oregon

July 21-25, '25 **Formal power series and algebraic combinatorics**, *Hokkaido University*, Sapporo, Japan

March 21-22, 2025 **Mid-Atlantic Algebra, Geometry, and Combinatorics (MAAGC) Workshop**, *CUNY Graduate Center in New York*

Jan. 20-24, '25 **Apprenticeship program in commutative algebra**, *Field's institute*, Toronto, Canada

July 22-26, '24 **Formal power series and algebraic combinatorics**, *Ruhr-Universität Bochum*, Bochum, Germany.

July 15-19, '24 **Summer school in algebraic combinatorics**, *Max Planck Institute*, Leipzig, Germany.

June 17-28, 2024 **Encuentro Colombiano de Combinatoria (ECCO)**, *Universidad del Cauca*, Popayan, Colombia

March 15-17, 2024 **Graduate Student Combinatorics Conference (GSCC)**, *Carnegie Mellon University*, Pittsburgh, PA.

June 12-22, '23 **Schubert summer school**, *University of Illinois at Urbana-Champaign*, Champaign, Illinois.

Apr. 28-30, '23 **Algebraic geometry northeastern series (AGNES)**, *Stony Brook University*, Stony Brook, NY.

Sep. 24, '22 **Route 81**, Queen's university

June 6-10, '22 **The Los Angeles Workshop on Representations and Geometry**, University of Southern California

May 6-8, '22 **Cornell Topology Festival**, Cornell university

Apr. 29-May 1, '22 **Derived categories and moduli spaces**, Cornell university

Organized

Fall 2022 **What is... seminar**, Cornell university, `https://pi.math.cornell.edu/m/event-list-p/whatis`

Awards/Scholarships

Awards

2015–2019 **Dean's honour list**, *University of Ottawa*

2018 **Student paper award**,
Department of mathematics and statistics, University of Ottawa
(Awarded for paper titled *Decomposing Frobenius Heisenberg categories*.)

External scholarships

2021 **Canada Graduate Scholarship - Doctoral**, *NSERC*
(Awarded for Sep. 2021 - Aug. 2024 session.) - Declined

2021 **Postgraduate Scholarship - Doctoral**, *NSERC*
(Awarded for Sep. 2021 - Aug. 2024 session.) - Accepted

2020 **Ontario graduate scholarship**, *Ontario government*
(Awarded for Sep. 2020 - Aug. 2021 session.) - Accepted

2019 **Canada graduate scholarship - M.Sc.**, *NSERC*
(Awarded for Sep. 2019 - Aug. 2020 session.) - Accepted

2019 **Ontario graduate scholarship**, *Ontario government*
(Awarded for Sep. 2019 - Aug. 2020 session.) - Declined

2019 **Undergraduate student research award**, *NSERC*

2018 **Undergraduate student research award**, *NSERC*

2017 **Undergraduate student research award**, *NSERC*

Internal scholarships

2021 **Admission fellowship**, *Cornell University*
(Awarded in 2021. Used during Aug. 2024-May 2025 session.)

2020 **Excellence scholarship - M.Sc.**, *University of Ottawa*
(Awarded for Sep. 2020 - Aug. 2021 session.)

2019 **Excellence scholarship - M.Sc.**, *University of Ottawa*
(Awarded for Sep. 2019 - Aug. 2020 session.)

2016 **Undergraduate research opportunity program**, *University of Ottawa*
2015 **Admission scholarship - B.Sc.**, *University of Ottawa*